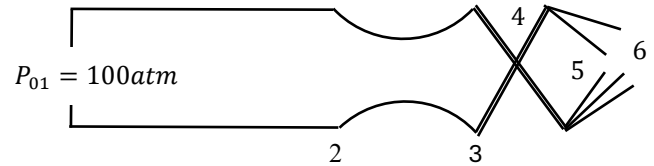


1- A converging-diverging nozzle is supplied by air with 100 atm pressure from a large combustion chamber. If the pressure at the nozzle exit is equal to 0.3 atm and the pressure in regions 4 and 6 is equal to ambient pressure, i.e., 1 atm:

- a- Find A_3/A^*
- b- Find M_3, M_4, M_5, M_6
- c- Find flow angle at region 4 and 6.



2. An ideal gas mixture at $T = 2000$ K has the following composition:

Species	H ₂	O ₂	H ₂ O	OH	O	H	CO	CO ₂
Mole fraction	0.086	0.019	0.326	0.060	0.012	0.030	0.319	0.148

The molar specific enthalpies and the constant pressure molar specific heats for the various species at $T = 2000$ K are listed in the table, below.

Species	H ₂	O ₂	H ₂ O	OH	O	H	CO	CO ₂
$\bar{h}$ (kJ/kmol)	52968	59169	-169040	92761	284909	253352	-53804	-302126
$\bar{c}_p$ (kJ/kmol-K)	34.236	37.788	51.143	34.635	20.819	20.786	36.721	60.433

- (a) Calculate, for the mixture, the molar mass ($\bar{M}_m$), the molar specific enthalpy ($\bar{h}_m$), the specific enthalpy (h_m), the specific gas constant (R_m) and the specific heat ratio (γ_m). {Partial Ans: $h_m = -4059$ kJ/kg, $R_m = 356.3$ J/kg-K}
- (b) The mixture is formed from the reaction of hydrocarbon (CH_{1.96}) and oxygen (O₂). Calculate the number of moles of hydrocarbon and the oxygen in the reactant mixture. From the result, calculate their mole fractions. { Partial Ans.: $X_{CH_{1.96}} = 0.4705, X_{O_2} = 0.5295$ }